TrES-5b

R-1193

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_211008` · created 2026-07-31T05:54:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459495.7141 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.002 +/- 0.054
Transit depth (Rp/Rs)^2	0.00055 +/- 0.025 [%]
Orbital Inclination (inc)	87.37 +/- 2.88
Airmass coefficient 1 (a1)	1.51 +/- 0.44
Airmass coefficient 2 (a2)	-0.25 +/- 0.25
Scatter in the residuals of the lightcurve fit is	31.57 %
Best Comparison Star	#1 - [547, 30]
Optimal Aperture	2.76
Optimal Annulus	3.25
Transit Duration (day)	0.072 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.002 ± 0.054 vs 0.142 (deviation 2.4σ)
Duration fit vs published	0.072 d vs 0.0758 d (deviation 0.3σ)
Mid-time O–C	-1.9 min
Depth SNR	0.0
Residual scatter	31.57 %

Light curve

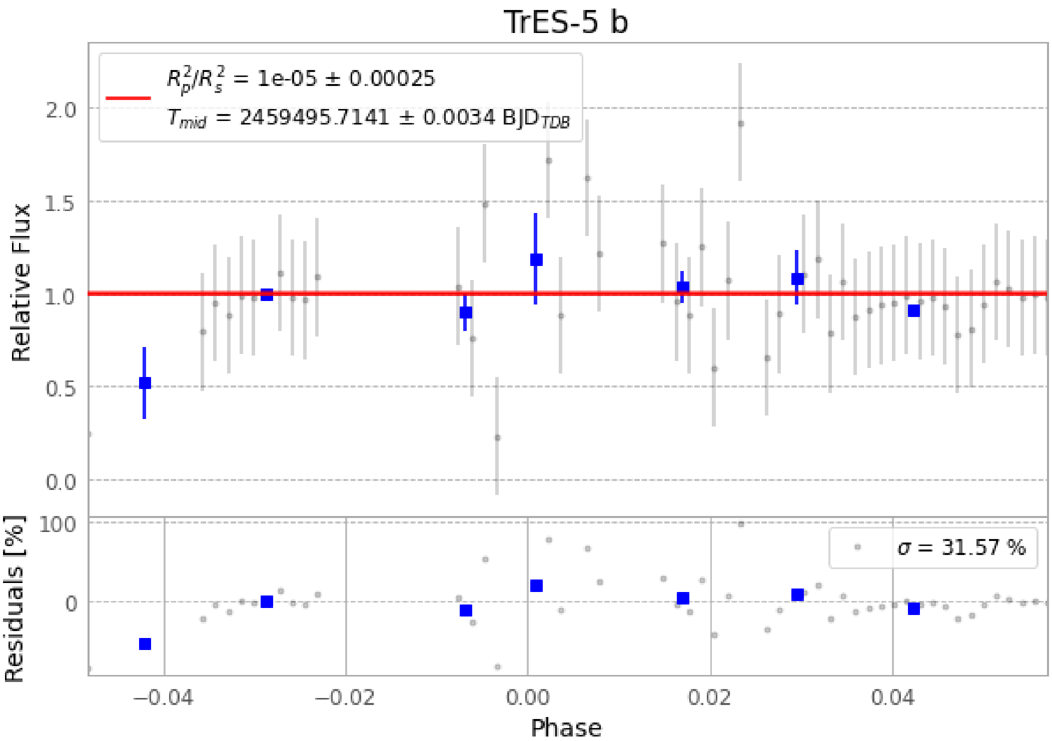

Figure 1 — FinalLightCurve_TrES-5 b_2021-10-07.png (SHA-256 c41f4ad83540218e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [458, 273], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 61c5b3549f9f59b4…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

76 FITS frames (50 MB), `TRES-5211008032112.FITS` → `TRES-5211008070612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-211008.log (3452e32a0a61…), FinalLightCurve_TrES-5 b_2021-10-07.png (c41f4ad83540…), FinalLightCurve_TrES-5 b_2021-10-07.csv (1e4eab1c1bd4…)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)